CLASS 9 — SCIENCE · FORMULA SHEET

Chapter 1: **Exploration: Entering the World of Chemistry** | Quick revision, one page

Nature of science

- Observation $\rightarrow$ question $\rightarrow$ hypothesis $\rightarrow$ experiment/testing $\rightarrow$ conclusion; results are shared and checked by other scientists.
- **Law**: describes a regular pattern in nature. **Theory**: well-tested explanation that makes testable predictions.
- **Model**: simplified representation that keeps what matters; models are revised when evidence disagrees with them.
- **Idealisation**: ignoring minor effects (e.g. frictionless surface, point mass) to study the main principle.

SI units

- Length: metre (m); mass: kilogram (kg); time: second (s); temperature: kelvin (K); amount of substance: mole.
- Prefixes: kilo (k) $= 10^{3}$, centi (c) $= 10^{-2}$, milli (m) $= 10^{-3}$, micro (μ) $= 10^{-6}$, nano (n) $= 10^{-9}$.
- $1\ \text{km} = 1000\ \text{m}$, $1\ \text{m} = 100\ \text{cm} = 1000\ \text{mm}$, $1\ \text{h} = 3600\ \text{s}$.

Unit conversion

- $1\ \text{km/h} = \dfrac{1000}{3600}\ \text{m/s} = \dfrac{5}{18}\ \text{m/s}$.
- km/h $\rightarrow$ m/s: multiply by $\dfrac{5}{18}$; m/s $\rightarrow$ km/h: multiply by $\dfrac{18}{5}$.
- Multiply by a conversion factor equal to 1, e.g. $\dfrac{1000\ \text{m}}{1\ \text{km}}$, so that the unwanted unit cancels.

Measurement and error

- Least count = smallest value that an instrument can measure (smallest scale division).
- Mean of readings $= \dfrac{\text{sum of readings}}{\text{number of readings}}$.
- Error = measured value $-$ true value; repeating and averaging reduces random error.
- Precision: closeness of repeated readings to each other. Accuracy: closeness to the true value.

Estimation and notation

- Scientific notation: $N = a \times 10^{n}$ with $1 \le a < 10$.
- Estimation gives an approximate value using round numbers; it is used to check whether an answer is sensible.
- The speed of light in vacuum is denoted by $c \approx 3 \times 10^{8}$ m/s.